Qwen

Governing the City at Scale: An Architectural Blueprint for Aletheion's Great_Functions Layer

The ALN Metaschema: Establishing a Non-Malformable Value Space

The foundational challenge in creating a resilient and trustworthy smart-city framework is ensuring that its core operational logic cannot be inadvertently or maliciously altered by human operators or developers. The concept of "Great_Functions" as a Kotlin-first governance layer is designed to solve this problem by establishing a rigid, machine-enforceable value space . This is achieved through the definition and enforcement of a minimalist but comprehensive ALN (Aletheion Language of Norms) metaschema for all governed decisions, coupled with a strict propagation standard for contextual data like `BirthSignId`. This architectural approach moves governance away from ad-hoc code constructs such as boolean flags or domain-specific enums, which are prone to drift and misinterpretation across different modules, and instead codifies it within a formal, shared grammar . By making every decision travel within a small, closed envelope whose contents are strictly typed and validated against this schema before any function call is permitted, the system creates an impenetrable barrier to value corruption .

The cornerstone of this non-malformable value space is the `ALE-TRUST-GOVERNED-DECISION-TX-001.aln` metaschema. This schema defines the immutable structure of the "Governed Decision Envelope," which serves as the sole output type consumed by clients, including Android applications and other backend services . Its purpose is to encapsulate not just the outcome of an action but also the complete provenance of that decision, thereby preventing access to unvetted internal states or optimization parameters . The proposed schema includes several critical fields. The `birthSignId` field anchors the decision to a specific territory, binding it to a unique bundle of lawful, ecological, and Indigenous norms, thus preventing the splicing of parameters from one jurisdiction to another . The `workflowId` identifies the specific engine or process that generated the decision, providing clear attribution . A critical component is the array of `alnNorms[]`, which explicitly lists the RightsAtoms, BioticTreaties, and other legal constructs that were evaluated during the pre-flight check, ensuring full transparency and

traceability . To maintain integrity, the envelope must also contain a cryptographic hash of the inputs and outputs of the workflow, allowing for verification of data integrity throughout the process . Finally, the `decisionOutcome` field contains the result of the authorization step, which can only be one of three explicit, non-probabilistic states: PERMIT, DEFER, or DENY, aligning with fail-closed principles where any ambiguity results in denial [1]. Great_Functions acts as the gatekeeper, refusing to emit any decision that does not conform precisely to this schema, effectively enforcing a contract for every automated action across the entire system .

To further solidify this value space, the system mandates that all values influencing a decision—be they limits, targets, thresholds, or allowed modes—must be scoped to a `BirthSignId` . This prevents humans from introducing unscoped literals or relying on user input that might ignore the treaty-bound properties of a given territory . This requires the establishment of a formal "Birth-Sign propagation standard." This standard dictates how the `BirthSignId` is passed through the system's request-response lifecycle. It would define a required field name, such as `x-birth-sign-id`, for inclusion in API headers or other communication protocols, and mandate a presence check at the earliest possible point of entry into the Great_Functions layer . Any service receiving a request without a valid `BirthSignId` would be rejected immediately, long before any business logic is executed. This ensures that a request to change a traffic light phase in a specific corridor is always evaluated against the set of norms associated with that corridor's `BirthSignId`, not some default or globally configured value. This practice is analogous to Zero Trust frameworks, which demand continuous verification of identity and context rather than assuming trust based on network location [9]. The integration of this standard with the CI/CD pipeline is crucial; builds could be configured to fail if a new module attempts to introduce governance flags that are not present in the master `ALE-GOV-RIGHTS-TREATY-GRAMMAR-001.aln` file, enforcing schema compliance at the development stage itself . This combination of a rigid ALN schema and a mandatory propagation standard transforms the system's value space from a mutable, human-managed construct into a static, machine-verifiable one, where proposals can be made but constraints cannot be reshaped outside the defined governance boundaries.

Field Name	Data Type	Description
`birthSignId`	String (UUID-like)	A unique identifier linking the decision to a specific geographic and legal zone, bound by its associated treaties and norms.
`alnNorms`	Array of Strings	A list of references to the specific RightsAtoms, BioticTreaties, and MicroTreaties that were evaluated during the Governance Preflight.
`workflowId`	String	A unique identifier for the specific Rust/ALN/Lua engine that processed the request and generated the decision.
`inputsHash`	String (SHA-3 Hash)	A cryptographic hash of the state snapshot and request payload received by the optimizer, ensuring input integrity.
`outputsHash`	String (SHA-3 Hash)	A cryptographic hash of the raw output from the optimization engine, before governance validation.
`decisionOutcome`	Enum	A single, definitive outcome from a finite set: PERMIT, DEFER, or DENY. Probabilistic outcomes are not allowed. [1]
`reasoning`	String (Optional)	A human-readable explanation for the `decisionOutcome`, detailing which norms were violated or why an action was deferred.
`signature`	String (EdDSA Signature)	A digital signature over the entire envelope's content, providing cryptographic proof of authorship and immutability.

This structured approach directly addresses the need for a rigid, machine-enforceable value space. By embedding minimal Googolswarm envelopes that include the `workflowId`, `BirthSignId`, and hashes of request/response, every citizen-facing action becomes fully traceable and auditable . The Kotlin SDK provides the hooks for building these envelopes but never implements consensus itself, keeping superpowers off the handset while still providing a tamper-evident record of intent . This design choice has profound implications for security and accountability. Every action taken by an application, from adjusting a heat budget to requesting more irrigation, generates a cryptographically signed and verifiable record. This record can be stored in an append-only log, creating a replayable, provenance-complete chain of decisions that can be used for post-hoc verification, compliance audits, and even counterfactual re-evaluation under new policy sets without having to re-run the original agent reasoning [1] . This architecture effectively turns the potential for human error or malice into a predictable and contained variable, ensuring that while humans can propose scenarios, they cannot fundamentally reshape the system's constraints, call paths, or treaty-bound properties outside the rigid envelope enforced by Great_Functions.

Nexus Mapping and Zone-Centric Orchestration for Cross-Workflow Interoperability

For a smart-city framework like Aletheion to scale to city-wide deployment, it must transcend the limitations of siloed, single-purpose workflows. The directive for Great_Functions is to act as a central nervous system capable of orchestrating simultaneous operations across potentially hundreds of distinct workflows, connecting disparate files and computational processes through a unified spatial and semantic model. This is achieved through two interconnected concepts: a cross-workflow nexus mapping system and a zone-centric zoning model. Together, these mechanisms create a dynamic, discoverable, and interoperable graph of all automated functions operating within the city, enabling intelligent routing of requests and collaborative problem-solving without requiring custom wiring for each new application or service .

The linchpin of this interoperability is the introduction of a shared Kotlin data model named `NexusCoordinate` . This coordinate system binds a physical or administrative place to a rich set of metadata that governs its operational logic. A `NexusCoordinate` object contains at least three key pieces of information: the `zoneId` (a unique identifier for a geographic tile or administrative area), the `BirthSignId` that defines its legal and ecological context, and a list of ERM domains it belongs to, such as water, thermal, mobility, or waste management . Crucially, it also links to the specific workflow IDs (e.g., `AWP_WATER_ALLOCATION`, `THERMAPHORA_HEAT_BUDGET`) that are responsible for governing that particular nexus . This simple yet powerful data structure transforms static locations into dynamic nodes in a larger operational graph. For example, when a citizen uses a heat-safety app to request cooler air in a specific corridor, the app does not need to know which backend service manages that corridor's temperature. Instead, it calls a Great_Functions registry method like `resolveNexus(corridorId)`, which returns the full `NexusCoordinate` for that location . This response reveals that the `THERMAPHORA_HEAT_BUDGET` workflow is active in that zone, allowing the app to then target its subsequent request to the correct engine. This decouples the client application from the backend implementation, a principle also seen in modern API design where semantic APIs expose entities and their relationships without exposing underlying data models [3] . The system allows any application view—whether for waste, heat, or mobility—to query, "what automations govern this tile?" through a standardized interface, dramatically simplifying the integration of new services and enabling emergent functionality .

Building upon the `NexusCoordinate`, the second pillar is the zone-centric zoning model, which unifies physical infrastructure with legal and governance frameworks. This

is implemented by binding Kotlin `ZoneDescriptor` objects directly to `BirthSignId`s . Each `ZoneDescriptor` becomes a comprehensive container for all relevant information about a territory. It holds the `BirthSignId`, defines its governance domains, and maintains explicit links to higher-level legal instruments such as BioticTreaties, Free, Prior, and Informed Consent (FPIC) flags, and local micro-treaties . This transforms a simple geographic zone into a legally and ecologically aware "service cell" . When an operation is initiated, Great_Functions can programmatically evaluate not only the current state model but also the treaty obligations embedded within the `ZoneDescriptor`. For instance, an action to extend store hours might be permissible in one zone but violate environmental noise agreements in another, a distinction that the `ZoneDescriptor` makes explicit. This is a sophisticated implementation of policy-as-code, where high-level legal constraints are translated into machine-executable rules that govern all actions within a specific territory. The enforcement is centralized within Great_Functions, which ensures that any Android action, regardless of its origin, is evaluated against the same ALN/Birth-Sign constraints that apply to backend Rust optimizers . This uniformity is critical for maintaining fairness and adherence to the city's foundational legal framework across all points of interaction.

To manage the complexity of hundreds of workflows and project future growth, the research directive specifies the need to detail the completion of the first 25 workflows while projecting the structure of the next 25 in GitHub-compatible YAML syntax. This is a pragmatic and essential strategy for achieving scalability, collaboration, and version control. YAML is an ideal choice for this task as it is a human-readable data serialization language widely used for configuration files and templates in modern DevOps practices, including Kubernetes and various CI/CD tools [5] [14] [27]. Adopting a template-based approach, similar to those used for generating Alloy concepts from OCCI templates [4], would ensure structural consistency across all new modules. A base YAML template would define the required structure for a compliant workflow, including placeholders for the `workflowId`, a list of ERM domains it interacts with, the entry points for the SMARTchain pattern, and references to its corresponding Rust/ALN core. Developers would then contribute new workflows by filling out this standardized YAML file, significantly lowering the barrier to entry and accelerating development cycles. This approach provides several key benefits: it ensures that every new workflow automatically adheres to the `GreatFunctionBoundary` contract enforced by Great_Functions; it makes the overall architecture more discoverable, as the structure of future workflows is predictable and can be programmatically parsed; and it provides a clear, version-controlled artifact for each new component of the city's automation system, facilitating auditing and maintenance. The table below illustrates a simplified YAML structure for a new workflow, serving as a blueprint for both completing the initial set and projecting the next 25.

```yaml
## Projected Workflow Structure for `AWP_WATER_ALLOCATION`
## This YAML file serves as a template for new Aletheion workflows,
## ensuring adherence to the Great_Functions boundary and SMARTchain patte
## Author: Aletheion Dev Team
## Version: 1.0

name: "Water Allocation Optimizer"
id: "AWP_WATER_ALLOCATION"
version: "1.2.0"

## Defines the primary ERM domains this workflow operates within.
## This enables the nexus-mapping system to discover its scope.
domains:
  - "water"
  - "thermal"

## Specifies the computational limits and resource requirements.
## Used for hardware-aware orchestration.
hardware_profile: "edge-moderate" # e.g., 'ultra-low-power', 'edge-moderat

## Defines the entry points for the six-stage SMARTchain pattern.
stages:
  ingest:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/ingest"
    description: "Receive real-time sensor data and state snapshots."
  state_model:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/state_model"
    description: "Update the internal state representation of the water gr
  optimization:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/optimization"
    description: "Execute the core optimization algorithm (Rust/ALN)."
  governance_preflight:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/preflight"
    description: "Validate the optimization output against ALN norms and B
  actuation:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/actuate"
    description: "Send commands to physical actuators (e.g., valve control
  trust_append:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/trust_append"
    description: "Prepare and submit the Governed Decision Envelope to the
```

```
  citizen_surface:
    endpoint: "/v1/workflow/AWP_WATER_ALLOCATION/surface"
    description: "Format and send the final decision back to the citizen-f

## Links to the external core that implements the business logic.
## This abstracts away the implementation language (e.g., Rust, ALN, Lua).
core_engine:
  language: "Rust"
  repository: "git@aletheion-core.org:engine-water-optimizer-v2.git"
  main_module: "src/lib.rs"

## Metadata for the CI/CD pipeline.
ci_cd:
  lint_rules:
    - "enforce-aln-schema-import: ALE-TRUST-GOVERNED-DECISION-TX-001.aln"
    - "require-birthsign-propagation"
```

By combining the `NexusCoordinate` for dynamic discovery, the `ZoneDescriptor` for contextual governance, and a standardized YAML projection for scalable development, Great_Functions establishes a robust foundation for cross-operational orchestration. This architecture allows the system to grow organically from an initial set of 25 workflows to hundreds, all interconnected and interoperable through a common semantic and spatial language.

Standardizing the SMARTchain Automation Pattern for Deterministic Action

To ensure that all automated actions within Aletheion are consistent, deterministic, and auditable, the framework must adopt a standardized workflow pattern. The six-stage SMARTchain pattern—Ingest → State-Model → Optimization → Governance Preflight → Actuation → Trust Append → Citizen Surface—is proposed as this canonical sequence . Great_Functions plays a pivotal role in enforcing this pattern by exposing its stages as callable entry points, thereby creating a mandatory, non-bypassable enforcement layer between a proposal's generation and its real-world execution [1]. This design directly addresses the user's requirement to prevent direct function calls to domain actuators and to standardize the "automation routine settlement" for Android operations . By sequencing the entire process—from pulling state from Layer2 to updating the UI—

through a shared `GreatFunctionRunner`, the system guarantees that every change request, whether from a citizen's app or a backend scheduler, undergoes the same rigorous governance checks .

The SMARTchain pattern is a formalized embodiment of the separation of concerns between proposing an action and authorizing its execution, a principle central to advanced security architectures like the Action Authorization Boundary (AAB) [1] . Each stage of the chain corresponds to a distinct logical step in the decision-making lifecycle. The **Ingest** and **State-Model** stages correspond to gathering the necessary context and representing it in a structured format, akin to preparing a Canonical Action Representation (CAR) [1] . The **Optimization** stage represents the agent's reasoning or planning phase, where it explores various possibilities to find an optimal solution. However, the critical innovation of the SMARTchain lies in the mandatory **Governance Preflight** stage. Before any value leaves Great_Functions, a preflight call is made to a shared, centralized compliance runtime that evaluates the proposed action against a suite of policies, including neurorights, FPIC, and BioticTreaties . This runtime is implemented as a separate, shared module, removing the possibility of per-app "shortcuts" around these fundamental constraints . Only if this preflight succeeds can the workflow proceed.

Following a successful preflight, the process continues to **Actuation**, where the command is sent to the physical world, and **Trust Append**, where the final decision is cryptographically sealed and recorded. This final stage is where the `Googolswarm-ready decision envelope` is created and submitted to the trust layer . As previously detailed, this envelope is a `ALE-TRUST-GOVERNED-DECISION-TX-001.aln` structure containing the `birthSignId`, applied `alnNorms`, and a cryptographic signature, ensuring full traceability for every citizen-facing action . The final stage, **Citizen Surface**, involves formatting this trusted outcome and delivering it back to the user interface, completing the feedback loop. The `GreatFunctionRunner` orchestrates this entire sequence, providing a single, standardized "automation handshake" that all domains—from water and heat to waste and Somaplex routes—must use . This uniformity is what enables true interoperability; because every action follows the same path, systems can reason about each other's intentions and collaborate safely. For example, an optimization in the `THERMAPHORA_HEAT_BUDGET` workflow can consult the `NexusCoordinate` to see that the `AWP_WATER_ALLOCATION` workflow is managing a nearby district, allowing for coordinated resource management at a city scale.

SMARTchain Stage	Corresponding AAB Concept	Description & Purpose	Great_Functions Enforcement
Ingest	Context Collection	Receive initial request and pull the latest state snapshot from a Layer2 source (e.g., REST/GraphQL API).	Provides a standardized interface to fetch state, ensuring all workflows start with a consistent view.
State-Model	Context Normalization	Update the internal state representation of the system based on the ingested data. [63] [65]	Abstracts the state model implementation, focusing on the public API for state updates.
Optimization	Proposal / Reasoning	Execute the core domain-specific optimization logic (e.g., in Rust/ALN) to generate a candidate action. [37] [41]	Wraps the external core engine, mediating the call and capturing its output for the next stage.
Governance Preflight	Authorization Evaluation	Validate the optimization output against ALN norms, BirthSign constraints, and centralized compliance checks (LexEthos, etc.).	Acts as the mandatory, non-bypassable authorization boundary. Rejects actions that fail preflight. [1]
Actuation	Execution	Send the authorized command to the physical or digital actuator (e.g., valve, route planner). [13]	Executes the command only after receiving a `PERMIT` decision from the preflight stage.
Trust Append	Logging & Provenance	Prepare a `GovernedDecisionEnvelope` with cryptographic proofs and submit it to the trust layer (Googolswarm).	Constructs and submits the immutable audit record, ensuring every action is traceable.
Citizen Surface	Feedback	Format the final decision outcome and deliver it back to the originating Android UI or API client.	Provides a standardized way to communicate results, showing the outcome and any reasons for rejection/ deratation.

This standardized pattern is not merely an organizational convenience; it is a core security and reliability feature. It creates a deterministic system where the path from proposal to execution is fixed and known. There are no hidden pathways or undocumented functions that can be called directly. This is particularly important in a city-scale deployment where the consequences of a rogue or improperly constrained action could be severe. By making the `Governed Decision Envelope` the only output type consumed by clients, the system enforces a principle of least disclosure, shielding them from sensitive internal weights, raw thresholds, and unvetted optimization outputs . This aligns perfectly with Zero Trust security principles, where every action must be verified and logged [9] . The resulting system is one where automation is not a black box but a transparent, auditable, and ultimately trustworthy process. The `GreatFunctionRunner` becomes the universal executor of this pattern, ensuring that no matter which of the hundreds of workflows is being invoked, the rules of engagement remain constant, fostering a secure and predictable environment for all urban operations.

Hardware-Aware Orchestration and Quantified Unit Proficiency Metrics

Deploying a system of Aletheion's ambition across an entire city necessitates moving beyond theoretical designs to concrete, hardware-aware planning. The user's directive for "low-power, city-grade, energy-conserving operation" demands a methodology for estimating hardware requirements and a mechanism for integrating hardware-aware, energy-conserving Googolswarm envelopes . This is achieved by defining a "quantified-unit proficiency" metric for each piece of infrastructure and leveraging this data for intelligent, resource-conscious orchestration. This approach acknowledges the heterogeneous nature of a smart city's edge computing landscape, which ranges from ultra-low-power microcontrollers to more powerful single-board computers, and seeks to match computational tasks to the most suitable hardware platform [88] [138].

The first step is to establish a standardized set of metrics to quantify the "proficiency" of any given hardware unit. These metrics must capture performance, energy consumption, and resource constraints, providing a holistic view of the device's capabilities. Based on research into Edge AI and IoT device benchmarking, a comprehensive proficiency profile should include the following:

1. **Performance Metrics:** This measures the raw computational throughput of the device. Key indicators include frames per second (FPS) for vision tasks, inference time in milliseconds (ms) for machine learning models, and overall processing power, often measured in FLOPs (floating-point operations per second) or CoreMark scores [57] [68] [85]. For example, a Jetson Nano might achieve up to 15 FPS for object detection, while simpler tasks might run faster on a Raspberry Pi [53] [85].
2. **Energy Consumption Metrics:** This is critical for the goal of energy conservation. It should measure power draw in different states (idle vs. active) in Watts (W), and total energy cost over time in kilowatt-hours (kWh) [94] [99]. Studies have shown that energy consumption varies significantly based on workload and timing behavior, making detailed profiling essential [117][137].
3. **Resource Constraint Metrics:** This captures the physical limitations of the device. It includes on-chip RAM/SRAM (in MB), storage capacity (in GB), and the specifications of the CPU/GPU/NPU (e.g., clock speed, number of cores, specialized acceleration units) [19] [52] [83].

By compiling these metrics into a "Hardware Proficiency Matrix," architects and developers gain a crucial reference tool for designing and deploying Aletheion's services. This matrix informs hardware-aware orchestration strategies, where computationally

intensive tasks, such as those involving complex simulations or large-scale data analysis, can be intelligently routed to more powerful devices like NVIDIA Jetson AGX Xavier platforms, which have demonstrated significant reductions in latency and memory use [86]. Conversely, simpler, repetitive tasks like real-time anomaly detection or signal processing can be offloaded to ultra-low-power MCUs based on the ARM Cortex-M series, which are designed for battery efficiency and operate within stringent power budgets [84] [88]. This tiered approach to computation is a hallmark of efficient Edge AI systems, balancing performance needs with energy realities [67].

Hardware Platform	Processor(s)	RAM	Typical Power Draw	Key Features & Use Case Suitability
STM32WB (with NPU)	Arm Cortex-M4, Cortex-M0+, NPU	Not Available	Low-Power	Highly integrated SoC with radio and dual MCUs. Ideal for RF-sensitive, low-power wireless sensor nodes and embedded control. [83] [101]
Raspberry Pi 4 Model B	1.5 GHz quad-core ARM Cortex-A72	2GB/4GB/ 8GB LPDDR4	~7-10W (under load)	General-purpose SBC. Suitable for lightweight analytics, data aggregation hubs, and less time-critical workflows. [115] [138]
Jetson Nano	Quad-core Arm Carmel CPU, 128-core Maxwell GPU	4GB LPDDR4	~5-10W	Entry-level AI computer. Capable of running DNNs for tasks like object detection and image classification at moderate frame rates (up to 15 FPS). [52] [85]
Jetson Xavier NX (AGX)	8-core Arm Carmel CPU, 384-core Volta GPU	8GB/16GB HBM2	~7-25W	High-performance edge AI. Suitable for complex, multi-modal AI workloads, large-scale DNN inference, and real-time autonomous processing. [86] [92]
Nordic nRF52832	Not Available	256 KB Flash, 64 KB RAM	Ultra-Low Power	Low-power MCU for Bluetooth Low Energy (BLE) applications. Ideal for simple sensor nodes and battery-powered devices where power draw is the primary constraint. [136]

With a quantified understanding of hardware capabilities, the integration of "energy-conserving Googolswarm envelopes" becomes a practical engineering challenge. These envelopes, which serve as the immutable records of decisions, must be designed to minimize the computational overhead of their creation and submission, especially on resource-constrained devices. This involves optimizing the cryptographic signing and hashing operations to use efficient algorithms (e.g., SHA-3, EdDSA) that are well-supported on embedded processors [91]. Furthermore, the submission process can be batched and scheduled during periods of low activity or when connected to a more powerful node, reducing the energy cost of network communication [51]. The entire SMARTchain process, from ingestion to trust appending, must be architected with these hardware constraints in mind. For instance, on a device with limited SRAM, model compression techniques like quantization may be necessary to keep memory usage low, a practice extensively studied for medical imaging and TinyML applications [98] [119]. The goal

is to create a system where the very act of proving a decision's validity (via the Googolswarm envelope) does not become a prohibitive drain on the device's energy or computational resources. This hardware-awareness extends to the software stack as well, leveraging official documentation and developer guidance on power modes and thermal management, as available for platforms like the Jetson Nano [57]. By meticulously estimating hardware requirements and designing the system—including its security mechanisms—to operate within those constraints, Aletheion can achieve its goal of being a city-grade, energy-conserving platform.

Ensuring Offline Security and Integrating Advanced Computational Paradigms

A smart-city framework deployed at city scale must be resilient, capable of functioning securely even when disconnected from central cloud services. This requirement for offline security preservation, combined with a forward-looking need to integrate advanced technologies like quantum-circuit learning algorithms and specialized monitoring systems like cyboquatic-nodes, defines the final frontier of Aletheion's architectural ambitions. The proposed solutions involve a hybrid architecture that leverages local computation and immutable logging for resilience, while simultaneously designing the system to be extensible enough to incorporate emerging paradigms for enhanced optimization and monitoring. This ensures the system is not only secure today but is also positioned to leverage tomorrow's breakthroughs in computing and sensing.

Offline security is primarily achieved through a decentralized approach to the SMARTchain process. Instead of relying on a constant, round-trip connection to a central authority, the architecture empowers individual edge nodes to perform the full sequence of operations locally [20]. When connectivity is lost, a node can continue to ingest data, model its local state, optimize its actions, and execute them autonomously. The critical steps for security—governance preflight and trust appending—are handled locally as well. The `GovernedDecisionEnvelope` is cryptographically signed by the local node, creating a tamper-evident record of the decision . This signed envelope is stored in a local, append-only log on the device itself. This local log serves as a temporary, secure ledger of all actions taken during the disconnection period. Once network connectivity is restored, these local logs can be synchronized with a central trust layer (like Googolswarm) in a batch process. This mechanism, a form of asynchronous consensus, ensures that the security and auditability guarantees of the system are maintained even in an offline state. This approach mitigates the attack surfaces inherent in edge

deployments of LLM agents, where a loss of connectivity could otherwise lead to uncoordinated or unauthorized behavior [16]. The use of Trusted Execution Environments (TEEs) and Secure Multi-Party Computation (SMPC) can further enhance this model by providing hardware-backed security for the signing and storage processes, protecting the integrity of the local logs even if the host device is compromised [82] [112].

Looking toward the future, the Aletheion architecture is designed to be extensible to advanced computational paradigms, most notably quantum computing. While still in its nascent stages, quantum computing offers significant potential for solving complex optimization problems that are intractable for classical computers, such as those found in power system management and logistics [37] [45]. The integration of quantum-circuit learning algorithms can be achieved through a hybrid quantum-classical model orchestrated by Great_Functions. In this model, Great_Functions acts as a dispatcher that identifies computationally hard problems within the SMARTchain—such as optimizing a large-scale energy grid or planning complex transportation routes—and routes them to a specialized quantum processor when one is available [41]. The Quantum Approximate Optimization Algorithm (QAOA), a prominent hybrid quantum-classical algorithm, provides a concrete pathway for this integration [41]. The quantum processor would provide a high-quality solution or a set of candidate solutions, which would then be fed back into the classical optimization stage of the SMARTchain for final refinement and validation against ALN norms. This hybrid approach leverages the strengths of both paradigms: the exponential search capability of quantum systems for exploring vast solution spaces, and the deterministic, rule-based validation of classical systems for ensuring safety and compliance. Research into optimizing quantum circuits using deep reinforcement learning further highlights the synergies between these fields, suggesting that AI can play a role in automating the design of quantum algorithms themselves [38] [42].

Finally, the architecture accommodates specialized monitoring systems like "cyboquatic-nodes." Although the term is not defined in the provided sources, we can infer its meaning from the context of pollution-reduction snapshots and internal monitoring . It likely refers to a conceptual fusion of cybernetic and aquatic systems, designed to monitor water quality, flow rates, ecosystem health, and related environmental indicators. In the Aletheion framework, these nodes would be treated as a specific type of sensor and would be integrated into the `AWP_WATER_ALLOCATION` or other relevant environmental management workflows. Each cyboquatic-node would report its findings through the standard Ingest and State-Model stages of the SMARTchain. The data from these nodes would contribute to the state model of the water ERM domain, influencing the Optimization stage—for example, by triggering adjustments to water release schedules from reservoirs to improve downstream ecological conditions [114]. The outputs

and snapshots from these nodes would be captured and appended to the trust layer via the Googolswarm envelope, providing a verifiable record of pollution reduction efforts and environmental stewardship. This integration demonstrates the flexibility of the SMARTchain pattern to incorporate diverse data sources and specialized hardware, enriching the state model and enabling more nuanced, data-driven governance. The entire system, therefore, forms a cohesive whole: a resilient, offline-capable architecture built on a foundation of formal schemas and standardized patterns, designed to be both operationally secure today and technologically adaptable for the challenges of tomorrow.

Reference

1. Faramesh: A Protocol-Agnostic Execution Control Plane for ... - arXiv https://arxiv.org/html/2601.17744v1
2. [PDF] AWS Well-Architected Framework https://docs.aws.amazon.com/pdfs/wellarchitected/2024-06-27/framework/wellarchitected-framework-2024-06-27.pdf
3. Semantic and modular orchestration of AI-driven digital twins for ... https://www.sciencedirect.com/science/article/pii/S2452414X25001827
4. [PDF] Inferring Models from Cloud APIs and Reasoning over Them https://theses.hal.science/tel-02016442v1/file/Challita-Thesis-final.pdf
5. Features - JetBrains Marketplace https://plugins.jetbrains.com/feature/
6. download as TXT - DistroWatch.com https://distrowatch.com/resource/freebsd/freebsd-12.3.txt
7. From AI for Science to Agentic Science: A Survey on Autonomous ... https://arxiv.org/html/2508.14111v2
8. Design of knowledge-based systems for automated deployment of ... https://www.researchgate.net/publication/344434775_Design_of_knowledge-based_systems_for_automated_deployment_of_building_management_services
9. Solutions - Zero Trust Frameworks Architecture Guide - Cisco https://www.cisco.com/c/en/us/solutions/collateral/enterprise/design-zone-security/zt-frameworks.html
10. Geopolitical ecologies of cloud capitalism: Territorial restructuring ... https://journals.sagepub.com/doi/10.1177/0308518X251369704
11. (Kotlin高级特性三）Kotlin密封类（Sealed Class）在何时比枚举更 ... https://www.cnblogs.com/gccbuaa/p/19229270

12. LinkedIn Learning Courses EN | PDF - Scribd https://de.scribd.com/document/708832052/LinkedIn-Learning-Courses-EN
13. 1. Introduction - arXiv.org https://arxiv.org/html/2603.04740v1
14. llm-ci-runner - PyPI https://pypi.org/project/llm-ci-runner/
15. Simple Index https://mirrors.aliyun.com/pypi/simple/
16. Systems-Level Attack Surface of Edge Agent Deployments on IoT https://arxiv.org/html/2602.22525v1
17. A portable diagnosis and distance localization approach for bearing ... https://www.sciencedirect.com/science/article/abs/pii/S1474034623003749
18. (PDF) Ultra-Low Power and Dependability for IoT Devices (Special ... https://www.researchgate.net/publication/312214220_Ultra-Low_Power_and_Dependability_for_IoT_Devices_Special_session_paper
19. Deploying LLM Transformer on Edge Computing Devices - MDPI https://www.mdpi.com/2673-2688/7/1/15
20. #ces2026 #ces2026 #ces #ai #edgeai #iot #lowpower | Mohit Agrawal https://www.linkedin.com/posts/mohitagrawal_ces2026-ces2026-ces-activity-7414731913812226048-NyAE
21. Assessing Digital Maturity in the Textile Sector: An Integrated ... - MDPI https://www.mdpi.com/2076-3387/16/3/135
22. [PDF] The heterogeneous band microstructure and mechanical ... - HAL https://hal.science/hal-04062122/document
23. [PDF] Building IBM Enterprise Content Management Solutions From End ... https://www.redbooks.ibm.com/redbooks/pdfs/sg248226.pdf
24. Workflow Control-Flow Patterns: A Revised View | Request PDF https://www.researchgate.net/publication/242388146_Workflow_Control-Flow_Patterns_A_Revised_View
25. [PDF] 北京邮电大学科技论文，专著（译著）统计2017 年度 https://kyy.bupt.edu.cn/__local/8/7C/A5/2FDC88F33CFA6DD5CE78850907F_CF3B77FF_8545C1.pdf?e=.pdf
26. [PDF] Implementation of a pipeline for collecting, ingesting and ... - Zenodo https://zenodo.org/records/15708394/files/Implementation%20of%20a%20pipeline%20for%20collecting,%20ingesting%20and%20transforming%20data%20into%20standard%20formats%20for%20the%20LAME%20FIB-SEM.pdf?download=1
27. Define drools conditions in configuration - rule engine - Stack Overflow https://stackoverflow.com/questions/76460123/define-drools-conditions-in-configuration
28. State-Change Learning for Prediction of Future Events in ... - arXiv https://arxiv.org/html/2510.12904v1

29. irsim — IR-SIM 2.9.1 documentation https://ir-sim.readthedocs.io/en/latest/api/irsim/index.html
30. (PDF) State-Change Learning for Prediction of Future Events in ... https://www.researchgate.net/publication/396517475_State-Change_Learning_for_Prediction_of_Future_Events_in_Endoscopic_Videos
31. Beyond green: sustainability in nuclear medicine practice - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12810880/
32. IoT Solution for Smart Cities' Pollution Monitoring and the Security ... https://www.mdpi.com/1424-8220/19/15/3401
33. Environment Pollution Analysis on Smart Cities Using Wireless ... https://www.researchgate.net/publication/367094347_Environment_Pollution_Analysis_on_Smart_Cities_Using_Wireless_Sensor_Networks
34. Artificial intelligence and IoT driven system architecture for ... https://www.sciencedirect.com/science/article/pii/S2665917424003714
35. [PDF] IoT- and AI-informed urban air quality models for vehicle pollution ... https://arxiv.org/pdf/2511.00187
36. AI-IoT Low-Cost Pollution-Monitoring Sensor Network to Assist ... https://www.mdpi.com/1424-8220/23/23/9585
37. Opportunities for quantum computing within net-zero power system ... https://www.sciencedirect.com/science/article/pii/S2542435124001557
38. Quantum circuit optimization with AlphaTensor - Nature https://www.nature.com/articles/s42256-025-01001-1
39. Optimization Strategies in Quantum Machine Learning - MDPI https://www.mdpi.com/2076-3417/15/8/4493
40. Interfacing Quantum Computing Systems with High-Performance ... https://arxiv.org/html/2509.06205v1
41. Quantum approximate optimization algorithm (QAOA) https://quantum.cloud.ibm.com/docs/tutorials/quantum-approximate-optimization-algorithm
42. Artificial intelligence for quantum computing - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC12673084/
43. Quantum computing: foundations, algorithms, and emerging ... https://www.frontiersin.org/journals/quantum-science-and-technology/articles/10.3389/frqst.2025.1723319/full
44. Exploring IoT and Blockchain: A Comprehensive Survey on Security ... https://www.mdpi.com/2504-2289/8/12/174

45. Blockchain technology in energy systems: A state − of − the − art review https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/blc2.12020
46. Vyoma commerce: a blockchain-based decentralized architecture to ... https://link.springer.com/article/10.1186/s43067-025-00245-6
47. tm2421409-15_f1 - none - 63.0725951s - SEC.gov https://www.sec.gov/Archives/edgar/data/1872195/000110465925069070/tm2421409-15_f1.htm
48. CompTIA SecurityX CAS-005 Exam Guide | PDF - Scribd https://www.scribd.com/document/946293400/CompTIA-SecurityX-CAS-005-Certification-Guide-sanet-st
49. Simple Index https://mirrors.aliyun.com/pypi/simple/?spm=a2c6h.13046898.publish-article.23.ee036ffaRjpFiW
50. Artificial Intelligence of Things for Next-Generation Predictive ... - MDPI https://www.mdpi.com/1424-8220/25/24/7636
51. A Low-Cost Energy-Efficient IoT Camera Trap Network for Remote ... https://www.mdpi.com/2079-9292/14/21/4266
52. AI-Based Embedded Framework for Cyber-Attack Detection Through ... https://www.mdpi.com/2076-3417/16/3/1416
53. Intrusion Detection on the Internet of Things - MDPI https://www.mdpi.com/2624-831X/7/1/16
54. (PDF) Edge AI for Real-Time Anomaly Detection in Smart Homes https://www.researchgate.net/publication/390924039_Edge_AI_for_Real-Time_Anomaly_Detection_in_Smart_Homes
55. AIP-Urban: Edge-Enabled Deep Learning Framework for Predictive ... https://www.mdpi.com/2079-8954/13/12/1117
56. Artificial intelligence of things for smart agriculture with optimal ... https://www.sciencedirect.com/science/article/pii/S0045790626001114
57. Benchmarking Deep Learning Models on NVIDIA Jetson Nano for ... https://www.researchgate.net/publication/392141157_Benchmarking_Deep_Learning_Models_on_NVIDIA_Jetson_Nano_for_Real-Time_Systems_An_Empirical_Investigation
58. Smart city energy efficient data privacy preservation protocol based ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11246532/
59. Digital Twins of Urban Air Quality: Opportunities and Challenges https://www.frontiersin.org/journals/sustainable-cities/articles/10.3389/frsc.2021.786563/full
60. Greening the Urban Landscape: Smart City Initiatives and Pollution ... https://www.mdpi.com/2079-8954/13/3/165

61. Smart City Initiatives and Pollution Reduction in China https://www.semanticscholar.org/paper/421a7992dc92ac7aa952e9af1976614cf33893e0
62. Hybrid approach for evaluating dynamic scheduling mechanisms in ... https://www.nature.com/articles/s41598-025-29635-6
63. Fast Real-Time Model Predictive Control for a Ball-on-Plate Process https://www.mdpi.com/1424-8220/21/12/3959
64. Pattern Recognition and Machine Learning (Information Science ... https://dl.acm.org/doi/10.5555/1162264
65. Demagnetisation fault analysis and diagnosis based on different ... https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/elp2.12519
66. Benchmarking Deep Learning Models for Object Detection on Edge ... https://arxiv.org/html/2409.16808v1
67. Edge Intelligence: A Review of Deep Neural Network Inference in ... https://www.mdpi.com/2079-9292/14/12/2495
68. Benchmarking Deep Neural Networks on Edge Devices, using ... https://outshift.cisco.com/blog/benchmarking-deep-neural-networks-on-edge
69. A Review of Embedded Machine Learning Based on Hardware ... https://pmc.ncbi.nlm.nih.gov/articles/PMC9959746/
70. Smart Cities: A Systematic Review of Emerging Technologies - MDPI https://www.mdpi.com/2624-6511/8/5/173
71. IoT-Driven Pollution Detection System for Indoor and Outdoor ... https://www.sciencedirect.com/org/science/article/pii/S1546221825012020
72. [PDF] Innovation landscape for smart electrification - IRENA https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2023/Jun/IRENA_Innovation_landscape_for_smart_electrification_2023.pdf
73. 6G—Enabling the New Smart City: A Survey - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC10490718/
74. Pollution Reduction, Informatization and Sustainable Urban ... - MDPI https://www.mdpi.com/2071-1050/15/13/10030?
75. CityPulse: Large Scale Data Analytics Framework for Smart Cities https://www.researchgate.net/publication/301240255_CityPulse_Large_Scale_Data_Analytics_Framework_for_Smart_Cities
76. [PDF] A Comprehensive Review of Quantum Circuit Optimization - arXiv https://arxiv.org/pdf/2408.08941
77. Optimal Quantum Circuit Design via Unitary Neural Networks https://www.researchgate.net/publication/383413422_Optimal_Quantum_Circuit_Design_via_Unitary_Neural_Networks

78. PSOC™ Control C3 Arm® Cortex®-M33 - Infineon Technologies https://www.infineon.com/products/microcontroller/32-bit-psoc-arm-cortex/32-bit-psoc-control-arm-cortex-m33-mcu
79. An Overview of Machine Learning within Embedded and ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8271867/
80. MCU Intelligent Upgrades: An Overview of AI-Enabled Low-Power ... https://www.mdpi.com/2079-9268/15/4/60
81. PSESV: A hybrid post-quantum encryption Framework with real-time ... https://www.sciencedirect.com/science/article/pii/S2090447925005179
82. Security in Embedded Devices: RTOS and Linux on ARM - LinkedIn https://www.linkedin.com/pulse/security-embedded-devices-rtos-linux-arm-gheorghe-prelipcean-upnlf
83. STMicroelectronics to boost AI at the edge with new NPU ... https://newsroom.st.com/media-center/press-item.html/p4665.html
84. A Systematic Review of State-of-the-Art TinyML Applications in ... https://ieeexplore.ieee.org/iel8/6287639/10820123/11266886.pdf
85. Benchmark Analysis of YOLO Performance on Edge Intelligence ... https://www.researchgate.net/publication/359689579_Benchmark_Analysis_of_YOLO_Performance_on_Edge_Intelligence_Devices
86. Onboard Optimization and Learning: A Survey - arXiv.org https://arxiv.org/html/2505.08793v2
87. Machine Learning at the Network Edge: A Survey https://dl.acm.org/doi/fullHtml/10.1145/3469029
88. Embedded Artificial Intelligence: A Comprehensive Literature Review https://www.mdpi.com/2079-9292/14/17/3468
89. On-line WSN SoC estimation using Gaussian Process Regression https://www.sciencedirect.com/science/article/pii/S1110016822001636
90. Integrity Guard 32 digital security technology https://www.infineon.com/technology/integrity-guard
91. Implementation and performance of post-quantum cryptography for ... https://link.springer.com/article/10.1007/s43926-025-00238-x
92. A Survey on Deep Learning Hardware Accelerators for ... - ACM https://dl.acm.org/doi/full/10.1145/3729215
93. [PDF] Intelligent World 2035 - Huawei https://www-file.huawei.com/admin/asset/v1/pro/view/8c64c0710ee04bee8e85385be5d944ad.pdf

94. (PDF) Energy and Sustainability - ResearchGate https://www.researchgate.net/publication/399128331_Energy_and_Sustainability
95. Electronics, Volume 15, Issue 3 (February-1 2026) – 226 articles https://www.mdpi.com/2079-9292/15/3
96. Hybrid Fusion Paradigm in Advanced Process Monitoring https://pubs.acs.org/doi/10.1021/acs.iecr.5c02759
97. Autonomous Indoor Drones and the Future of Objective Property ... https://arxiv.org/html/2508.01965v1
98. Tiny Machine Learning and On-Device Inference: A Survey ... - MDPI https://www.mdpi.com/1424-8220/25/10/3191
99. On resource consumption of machine learning in communications ... https://www.sciencedirect.com/science/article/pii/S1389128625005675
100. Google Cloud's LiteRT boosts TinyML for IoT devices - LinkedIn https://www.linkedin.com/posts/samiksha-yadav-002b63259_tinyml-litert-edgeai-activity-7369969424935624704-TbDW
101. STMicroelectronics Accelerates Wireless Product Development with ... https://newsroom.st.com/media-center/press-item.html/p4387.html
102. The neurobench framework for benchmarking neuromorphic ... https://www.nature.com/articles/s41467-025-56739-4
103. (PDF) Semantic Physics: A Framework for Structurally Stable ... https://www.researchgate.net/publication/396416842_Semantic_Physics_A_Framework_for_Structurally_Stable_Artificial_Intelligence_A_Proposal_for_Self-Regulating_AI_Architectures_Based_on_Semantic_Field_Theory
104. (PDF) SEMANTIC PHYSICS: FROM PHENOMENOLOGICAL ... https://www.researchgate.net/publication/397543573_SEMANTIC_PHYSICS_FROM_PHENOMENOLOGICAL_FRAMEWORK_TO_FUNDAMENTAL_ONTOLOGY
105. (PDF) PROTO-∇: The Convergence of Difference - ResearchGate https://www.researchgate.net/publication/398902177_PROTO-_The_Convergence_of_Difference
106. Simple Index http://mirrors.aliyun.com/pypi/simple/?spm=a2c4e.11153940.0.0.166e245d6Gtam5
107. (PDF) From Folding to Reasoning: Spectral Bifurcations and Phase ... https://www.researchgate.net/publication/396033792_From_Folding_to_Reasoning_Spectral_Bifurcations_and_Phase_Transitions_in_Typed_Semantic_Graphs

108. (PDF) Gravitas Semantica: A Theory of Held Tension - ResearchGate https://www.researchgate.net/publication/398123589_GRAVITAS_SEMANTICA_A_Theory_of_Held_Tension
109. Kotlin Value Class 全面解析：类型安全与零开销封装原创 - CSDN博客 https://blog.csdn.net/szhupeng/article/details/152219566
110. [PDF] Real-Time Machine Learning for Embedded Anomaly Detection - arXiv https://arxiv.org/pdf/2512.19383
111. Superconducting quantum computers: who is leading the future? https://link.springer.com/article/10.1140/epjqt/s40507-025-00405-7
112. Hybrid Federated Learning Architecture for Smart Urban Computing https://dl.acm.org/doi/fullHtml/10.1145/3467956
113. [PDF] Intelligent World 2030 - Huawei https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/intelligent_world_2030_2024_en.pdf?la=en
114. Appl. Sci., Volume 16, Issue 4 (February-2 2026) – 491 articles - MDPI https://www.mdpi.com/2076-3417/16/4
115. [PDF] Enhancing IoT Security using NFV-based Edge Traffic Analysis - arXiv https://arxiv.org/pdf/1805.10815
116. Edge Computing on IoT for Machine Signal Processing and Fault ... https://www.researchgate.net/publication/367454695_Edge_Computing_on_IoT_for_Machine_Signal_Processing_and_Fault_Diagnosis_A_Review
117. [PDF] Statistical models for the optimization of energy consumption in ... https://inria.hal.science/tel-04999351v2/file/141474_WEHAIBA_EL_KHAZEN_2024_archivage.pdf
118. [PDF] Charting Disruption 2023 - Global X ETFs https://assets.globalxetfs.com/files/ChartingDisruption2023.pdf
119. Quantization of Deep Neural Networks for Medical Image Analysis https://www.mdpi.com/2227-7080/14/1/76
120. [PDF] Fault Diagnosis Using Artificial Intelligence Techniques and Fault ... https://theses.hal.science/tel-05458506v1/file/143818_WU_2025_archivage.pdf
121. [PDF] Energy and AI - Microsoft .NET https://iea.blob.core.windows.net/assets/dd7c2387-2f60-4b60-8c5f-6563b6aa1e4c/EnergyandAI.pdf
122. Analysis of various techniques for ECG signal in healthcare, past ... https://www.sciencedirect.com/science/article/pii/S2667099223000191
123. Reinforcement Learning: An Introduction | Guide books https://dl.acm.org/doi/book/10.5555/3312046

124. QORECHAIN Quantum Safe AI Optimized Interchain Architecture https://www.academia.edu/144643193/QORECHAIN_Quantum_Safe_AI_Optimized_Interchain_Architecture
125. Complex, Intelligent and Software Intensive Systems - Springer Link https://link.springer.com/content/pdf/10.1007/978-3-031-96099-4.pdf
126. Advanced Network Technologies and Computational Intelligence https://link.springer.com/content/pdf/10.1007/978-3-031-86069-0.pdf
127. Intelligent Signal Processing and RF Energy Harvesting for State of ... https://link.springer.com/content/pdf/10.1007/978-981-99-8771-9.pdf
128. Advanced Information Networking and Applications - Springer https://link.springer.com/content/pdf/10.1007/978-3-031-57853-3.pdf
129. Intelligence of Things: Technologies and Applications - Springer https://link.springer.com/content/pdf/10.1007/978-3-031-46749-3.pdf
130. Innovations in Cybersecurity and Data Science - Springer Link https://link.springer.com/content/pdf/10.1007/978-981-97-5791-6.pdf
131. [PDF] Intelligent World 2030 - Huawei https://www-file.huawei.com/admin/asset/v1/pro/view/d2c1c28eeba24f4ca7bdf0022805a1dc.pdf
132. [XLS] China https://www.nsfc.gov.cn/Portals/0/fj/fj20220316_02.xlsx
133. Appl. Sci., Volume 15, Issue 20 (October-2 2025) – 439 articles - MDPI https://www.mdpi.com/2076-3417/15/20
134. [PDF] SIMATIC S7-1200 G2 Programmable Logic Controller - Support https://support.industry.siemens.com/dl/files/011/109972011/att_1308257/v1/S71200_G2_system_manual_en-US.pdf
135. Reduce RAM usage? - Jetson Nano - NVIDIA Developer Forums https://forums.developer.nvidia.com/t/reduce-ram-usage/84276
136. Minimizing the power draw using System OFF and RAM retention https://devzone.nordicsemi.com/f/nordic-q-a/40351/minimizing-the-power-draw-using-system-off-and-ram-retention
137. Energy-Saving Techniques for Raspberry PI | Request PDF https://www.researchgate.net/publication/395210335_Energy-Saving_Techniques_for_Raspberry_PI
138. Explain the differences and use cases for a Raspberry Pi vs Arduino ... https://dev.to/rickssss/explain-the-differences-and-use-cases-for-a-raspberry-pi-vs-arduino-like-im-five-but-smart-2ogn
139. [PDF] COSMOPlat IoT Technology Co., Ltd. 卡奥斯物聯科技股份有限公司 https://www1.hkexnews.hk/app/sehk/2026/108177/documents/sehk26013002645.pdf

140. Iot Esz 2025 | PDF - Scribd https://www.scribd.com/document/931805218/Iot-Esz-2025